

From Tanner Codes to Good Quantum LDPC Codes

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Abstract

Low-density parity-check (LDPC) codes are classical error-correcting codes whose parity constraints remain local as the block length grows. Classical expander codes show that this locality is compatible with both constant rate and linear distance. The quantum analogue is more subtle: a CSS code requires two sparse parity-check matrices H_X, H_Z satisfying $H_X H_Z^T = 0$, and its minimum distance is measured modulo stabilizers. Hypergraph-product codes give constant-rate qLDPC codes with distance of order $\sqrt{n}$. The 2021 breakthrough of Panteleev and Kalachev used lifted products over nonabelian group algebras, together with local product expansion and global Cayley expansion, to construct explicit qLDPC codes with constant rate and linear distance. This note reviews the classical Tanner-code paradigm, the CSS formalism, hypergraph products, the Panteleev–Kalachev construction, and several developments after the qLDPC conjecture was resolved.

1 Classical linear codes

A binary linear code of block length n is a subspace

$$C \subseteq \mathbb{F}_2^n.$$

If $\dim C = k$, then the code stores k information bits in n physical bits. Its minimum distance is

$$d(C) = \min_{0 \neq c \in C} \text{wt}(c).$$

Thus a classical linear code has parameters $[n, k, d]$.

Equivalently, a linear code can be specified by a parity-check matrix

$$H \in \mathbb{F}_2^{m \times n}, \quad C = \ker H.$$

If a codeword $c \in C$ is corrupted by an error e , the received word is $y = c + e$, and its syndrome is

$$Hy = H(c + e) = He.$$

The syndrome eliminates the unknown codeword and retains information about the error.

Two asymptotic parameters are fundamental:

$$R = \frac{k}{n}, \quad \delta = \frac{d}{n}.$$

A family is *asymptotically good* if both R and δ are bounded away from zero. It is *low-density parity-check* (LDPC) if the row and column weights of H are bounded independently of n . Thus every check touches only $O(1)$ symbols and every symbol participates in only $O(1)$ checks.

Simple examples illustrate the tension between rate and distance. The repetition code

$$0 \mapsto 0000, \quad 1 \mapsto 1111$$

has large relative distance but poor rate when the repetition length grows. The single-parity-check code

$$(m_1, \dots, m_k) \mapsto (m_1, \dots, m_k, m_1 + \dots + m_k)$$

has rate approaching one but minimum distance only 2. The classical goal is to achieve positive rate and positive relative distance simultaneously, ideally with sparse checks.

2 Tanner codes and expansion

Tanner codes provide a particularly transparent construction of good classical LDPC codes. Let

$$\Gamma = (L \sqcup R, E)$$

be a D -regular bipartite graph with

$$|L| = |R| = m, \quad |E| = Dm.$$

Place one code symbol on every edge. Choose a fixed local code

$$C_0 \subseteq \mathbb{F}_2^D$$

with parameters $[D, k_0, d_0]$. For a global word $x \in \mathbb{F}_2^E$ and a vertex $v \in L \sqcup R$, let

$$x|_{E(v)} \in \mathbb{F}_2^D$$

denote the restriction to the D incident edges. The Tanner code is

$$T(\Gamma, C_0) = \{x \in \mathbb{F}_2^E : x|_{E(v)} \in C_0 \text{ for every } v \in L \sqcup R\}.$$

If $C_0 = \ker h_0$, the global parity-check matrix is obtained by placing a copy of h_0 at every vertex. Since D and h_0 are fixed, the global family is LDPC.

The rate is already visible from constraint counting. There are Dm edge symbols and each of the $2m$ vertices imposes at most $D - k_0$ independent constraints. Hence the global dimension K satisfies

$$K \geq Dm - 2m(D - k_0) = m(2k_0 - D),$$

so

$$\frac{K}{Dm} \geq 2\frac{k_0}{D} - 1.$$

In particular, if $k_0/D > 1/2$, the Tanner code has positive constant rate.

The distance is where graph expansion enters. Let

$$x = \mathbf{1}_F \in T(\Gamma, C_0), \quad F \subseteq E,$$

be a nonzero codeword. Define the active vertices

$$S = \{u \in L : E(u) \cap F \neq \emptyset\}, \quad T' = \{v \in R : E(v) \cap F \neq \emptyset\}.$$

At each active vertex the local restriction is a nonzero word of C_0 , so

$$|F| \geq d_0|S|, \quad |F| \geq d_0|T'|.$$

Therefore

$$|F| \geq d_0\sqrt{|S||T'|}. \quad (1)$$

Let λ be the second singular value of the bipartite adjacency matrix. The expander mixing lemma gives

$$e(S, T') \leq \frac{D}{m}|S||T'| + \lambda\sqrt{|S||T'|}.$$

Since $F \subseteq E(S, T')$,

$$|F| \leq \frac{D}{m}|S||T'| + \lambda\sqrt{|S||T'|}.$$

Together with (1), this implies

$$\sqrt{|S||T'|} \geq \frac{d_0 - \lambda}{D}m,$$

and hence

$$|F| \geq \frac{d_0(d_0 - \lambda)}{D}m.$$

Since the block length is $N = Dm$,

$$\frac{d(T(\Gamma, C_0))}{N} \geq \frac{d_0}{D} \left(\frac{d_0}{D} - \frac{\lambda}{D} \right).$$

Thus if the local relative distance $\delta_0 = d_0/D$ satisfies $\delta_0 > \lambda/D$, the global Tanner code has linear distance.

The key classical mechanism is therefore

local distance + global graph expansion $\implies$ linear global distance.

3 Quantum codes and CSS codes

The state space of n qubits is

$$\mathcal{H}_n = (\mathbb{C}^2)^{\otimes n}.$$

Definition 1. An $[[n, k, d]]$ quantum code is a 2^k -dimensional subspace $Q \subseteq \mathcal{H}_n$ that detects every error supported on fewer than d qubits.

Let P_Q be the orthogonal projector onto Q . An operator E is detectable by Q if

$$P_Q E P_Q = \alpha_E P_Q$$

for some scalar α_E . The distance is the smallest weight of an error that is not detectable. Since the Pauli operators form a basis for all qubit operators, it is enough to analyze Pauli errors.

For one qubit the Pauli operators are

$$I, \quad X, \quad Z, \quad Y = iXZ.$$

Ignoring a global phase, every n -qubit Pauli operator can be written as

$$X^a Z^b, \quad a, b \in \mathbb{F}_2^n.$$

Thus Pauli errors are represented by two binary vectors: one records the X -part and the other the Z -part.

A stabilizer code is the common $+1$ -eigenspace of an abelian subgroup of the Pauli group not containing $-I$. CSS codes are the special case in which the stabilizer generators are separated into X -type and Z -type operators. Let

$$H_X \in \mathbb{F}_2^{m_X \times n}, \quad H_Z \in \mathbb{F}_2^{m_Z \times n}.$$

For each row x of H_X , use the stabilizer X^x , and for each row z of H_Z , use Z^z . Since

$$X^x Z^z = (-1)^{x \cdot z} Z^z X^x,$$

all stabilizers commute exactly when

$$H_X H_Z^\top = 0.$$

The code space is

$$Q = \{|\psi\rangle : X^x|\psi\rangle = |\psi\rangle \ \forall x \in \text{row}(H_X), \ Z^z|\psi\rangle = |\psi\rangle \ \forall z \in \text{row}(H_Z)\}.$$

Its number of logical qubits is

$$k = n - \text{rank } H_X - \text{rank } H_Z.$$

Quantum distance is measured modulo stabilizers. An undetected Z -type operator satisfies $H_X z = 0$, but it acts trivially on the code space if $z \in \text{row}(H_Z)$. Hence

$$d_Z = \min\{\text{wt}(z) : z \in \ker H_X \setminus \text{row}(H_Z)\},$$

and similarly

$$d_X = \min\{\text{wt}(x) : x \in \ker H_Z \setminus \text{row}(H_X)\}.$$

The quantum distance is

$$d = \min\{d_X, d_Z\}.$$

A CSS family is qLDPC if both H_X and H_Z have bounded row and column weights.

The Calderbank–Shor–Steane framework therefore turns the quantum coding problem into a binary one: construct two sparse classical constraint systems that remain large-distance after quotienting by stabilizers and that satisfy the extra orthogonality condition $H_X H_Z^\top = 0$.

4 The hypergraph product

Tillich and Zémor gave a canonical way to enforce the CSS commutation condition. Let

$$A \in \mathbb{F}_2^{m_A \times n_A}, \quad B \in \mathbb{F}_2^{m_B \times n_B}$$

be classical parity-check matrices. Define

$$H_X = \begin{pmatrix} A \otimes I_{n_B} & I_{m_A} \otimes B^\top \end{pmatrix},$$

and

$$H_Z = \begin{pmatrix} I_{n_A} \otimes B & A^\top \otimes I_{m_B} \end{pmatrix}.$$

Then

$$\begin{aligned} H_X H_Z^\top &= A \otimes B^\top + A \otimes B^\top \\ &= 0 \end{aligned}$$

over $\mathbb{F}_2$. If A and B are LDPC matrices, then the resulting CSS code is also qLDPC.

The number of qubits is

$$N = n_A n_B + m_A m_B.$$

For appropriate asymptotically good classical LDPC constituents of comparable length L , Tillich and Zémor obtain positive-rate qLDPC codes with minimum distance proportional to L . Since $N = \Theta(L^2)$,

$$D = \Theta(L) = \Theta(\sqrt{N}).$$

Thus the hypergraph product solves the commutation and sparsity problems, but its most natural parameter regime has only square-root distance.

5 The Panteleev–Kalachev breakthrough

Panteleev and Kalachev broke the square-root barrier by combining product constructions with a large nonabelian group symmetry.

Theorem 1 (Panteleev–Kalachev, 2021; informal form). *For every fixed $R \in (0, 1)$ and every finite field $\mathbb{F}_q$, there exists an explicit family of quantum LDPC codes with parameters*

$$[[n, k, d]]_q, \quad k \geq Rn, \quad d = \Theta(n),$$

as $n \rightarrow \infty$.

The algebraic mechanism is the lifted product. Suppose two classical codes are invariant under a free action of a finite group G on their coordinates and checks. Let

$$R = \mathbb{F}_q[G]$$

be the group algebra. Their parity-check maps can be written compactly as

$$A \in R^{m_a \times n_a}, \quad B \in R^{m_b \times n_b},$$

where m_a, n_a, m_b, n_b are constant while $|G|$ grows.

The lifted product is the tensor product over R . It gives a three-term sequence

$$R^{n_a n_b} \xrightarrow{\partial_2} R^{n_a m_b} \oplus R^{m_a n_b} \xrightarrow{\partial_1} R^{m_a m_b},$$

where, up to the conventional ordering of factors,

$$\partial_2 = \begin{pmatrix} A \otimes_R I \\ -I \otimes_R B \end{pmatrix}, \quad \partial_1 = \begin{pmatrix} I \otimes_R B & A \otimes_R I \end{pmatrix}.$$

The defining identity is

$$\partial_1 \partial_2 = A \otimes_R B - A \otimes_R B = 0.$$

This gives a CSS code by taking $H_X = \partial_1$ and $H_Z = \partial_2^\top$, with the appropriate left/right module adjoint in the nonabelian setting.

The block-length saving is captured by

$$R \otimes_R R \cong R.$$

The middle space contains only a constant number of copies of R , so its dimension over $\mathbb{F}_q$ is

$$N = |G|(n_a m_b + m_a n_b) = \Theta(|G|).$$

An ordinary tensor product over $\mathbb{F}_q$ would instead retain two independent group coordinates and have size $\Theta(|G|^2)$.

Linear block length alone does not imply linear quantum distance. This is the main technical difficulty. Panteleev and Kalachev instantiate the lifted product using Tanner codes built from highly symmetric Cayley expanders. The two constant-size local codes are selected to satisfy a stronger two-dimensional robustness property called *product expansion*. Informally, a local $w \times w$ pattern built from codeword-columns and codeword-rows cannot become very sparse merely through cancellation. In a suitable reduced decomposition one has, schematically,

$$\text{wt}(M) \gtrsim w(\#\text{active codeword-columns} + \#\text{active codeword-rows}).$$

This is the product analogue of the local-distance property in an ordinary Tanner code. Their distance proof then has the same local-to-global structure:

$\text{local product expansion} + \text{global Cayley expansion} \implies d = \Theta(n).$

The lifted product solves the size problem, while product expansion together with Cayley expansion solves the distance problem.

6 Developments after the qLDPC conjecture

The 2021 result settled the structural qLDPC conjecture, but subsequent work has separated several additional goals: simpler constructions, efficient decoding, more explicit graph-theoretic ingredients, and low-complexity quantum encoding.

Leverrier and Zémor introduced *quantum Tanner codes* in 2022. Their construction places classical Tanner codes on a left–right Cayley complex and gives a cleaner Tanner-style realization of asymptotically good qLDPC codes. It makes especially transparent the persistence of the classical philosophy: strong local constraints plus expansion can yield large global quantum distance, provided the two constraint systems satisfy the required CSS compatibility.

A second 2022 milestone was the work of Dinur, Hsieh, Lin, and Vidick. They constructed an explicit family of good qLDPC codes with linear-time decoders. Their codes are based on a left–right Cayley square complex and on a robust tensor-product property of constant-size local codes. Thus one can have simultaneously

$$k = \Theta(n), \quad d = \Theta(n), \quad O(1)\text{-weight checks},$$

together with an efficient classical syndrome decoder for a linear fraction of adversarial errors.

A related route due to Lin and Hsieh showed that suitable two-sided lossless expanders with a free group action would yield good qLDPC codes with linear-time decoding. At the time the required explicit graph family was not known. In 2025, Hsieh, Lubotzky, Mohanty, Reiner, and Zhang constructed the first explicit constant-degree lossless vertex expanders. Their graphs also admit the required free group action and therefore instantiate new qLDPC families through the Lin–Hsieh framework. The significance of this result is not that it was the first explicit good qLDPC construction, but that it completed a particularly clean lossless-expander route while solving an independent graph-theoretic problem.

Wills, Lin, Zhang, and Hsieh studied a different complexity axis in 2026. They constructed asymptotically good quantum codes whose encoding and unencoding circuits have logarithmic depth and linear total gate complexity, together with fast classical decoding; they also give explicit families with linear total complexity for encoding, unencoding, and decoding. These codes are aimed at the channel-capacity setting, in which encoding and unencoding are assumed ideal, and they are not asserted to be qLDPC. Hence this result does not supersede the qLDPC constructions. Instead, it raises the natural question of how many desirable structural and algorithmic properties can be combined in one family.

The modern landscape can therefore be summarized as follows:

good qLDPC existence: solved,
 good qLDPC with linear-time syndrome decoding: known,
 good quantum codes with linear-size encoding/unencoding: known,
 simultaneous optimization of locality, encoding, decoding,
 fault tolerance, logical operations, and local testability: active.

From a combinatorial perspective, the central questions have shifted from existence toward stronger notions of expansion, smaller locality, robustness, and compatibility with additional algorithmic structure.

7 Conclusion

The history of qLDPC codes mirrors classical expander coding, but with an essential quantum obstruction: the two families of parity checks must commute, and logical distance is measured modulo stabilizers. Hypergraph products preserve sparsity and enforce commutation, but naturally give a square-root-distance scaling. Panteleev and Kalachev used a lifted product over a nonabelian group algebra to remove one factor of the product size, then used local product expansion and global Cayley expansion to prove linear quantum distance. This resolved the qLDPC conjecture. Later work has focused on cleaner Tanner-style constructions, fast decoding, explicit lossless expanders, and fast encoding, turning the subject from an existence problem into a broader study of which structural and algorithmic properties can coexist.

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